Credit Card Fraud Detection System

Project Explanation Document

---

1. Introduction

With the rapid growth of online payments and digital banking, credit card usage has increased significantly. However, this growth has also increased the number of fraudulent transactions. Fraudulent transactions can cause financial loss to both customers and financial institutions.

To solve this issue, machine learning techniques can be used to automatically analyse transaction data and detect suspicious activities.

In this project, a **Credit Card Fraud Detection System** is developed using the **Random Forest Machine Learning algorithm**. The system analyses transaction details and predicts whether a transaction is **normal or fraudulent**.

2. Real Life (Day-to-Day) Example

Consider a person who regularly uses their credit card for daily purchases.

Normal Transaction Behaviour

Feature	Example
Purchase Amount	₹200 – ₹3000
Location	Same city
Transaction Type	Grocery purchase
Time	Daytime

Now suppose a transaction occurs:

Suspicious Transaction Behaviour

Feature	Example
Purchase Amount	₹80,000
Location	Different city
Transaction Type	Refund or unknown merchant

Feature	Example
Time	Midnight

Such unusual patterns can indicate **fraudulent activity**.

The machine learning system analyzes thousands of past transactions and learns these patterns. When a new transaction occurs, it predicts whether it is **safe or fraudulent**.

3. Dataset Description

The dataset used in this project contains **100,000 transaction records**.

Each record includes information about a credit card transaction.

Dataset Structure

Column Name	Description
TransactionID	Unique ID for each transaction
TransactionDate	Date and time of the transaction
Amount	Transaction amount
MerchantID	Unique ID of the merchant
TransactionType	Type of transaction (purchase/refund)
Location	Location where the transaction occurred
IsFraud	Target variable (0 = Normal, 1 = Fraud)

Dataset Size

Parameter	Value
Total Transactions	100,000
Normal Transactions	99,000
Fraud Transactions	1,000

This shows that **fraud transactions are very rare**, which creates an **imbalanced dataset problem**.

4. Data Imbalance Problem

Fraud detection datasets usually contain many normal transactions but very few fraud transactions.

Original Dataset Distribution

Transaction Type	Count
Normal	99,000
Fraud	1,000

If the machine learning model is trained directly on this dataset, it may become biased toward predicting normal transactions.

To solve this problem, the **SMOTE (Synthetic Minority Oversampling Technique)** method is used.

5. SMOTE Data Balancing

SMOTE generates synthetic samples of the minority class (fraud transactions) to balance the dataset.

Training Data Before Balancing

Transaction Type	Count
Normal	79,200
Fraud	800

Training Data After SMOTE Balancing

Transaction Type	Count
Normal	79,200
Fraud	79,200

After balancing, both classes have equal representation, allowing the model to learn fraud patterns better.

6. Data Pre-processing

Before training the machine learning model, several pre-processing steps are performed.

Pre-processing Steps

Step	Description
Dataset Loading	Transaction dataset is loaded into the system
Data Inspection	Dataset structure and columns are checked
Feature Separation	Input features and target variable are separated
Non-Numerical Removal	Text columns are removed for model training
Feature Scaling	Numerical features are normalized using StandardScaler

Feature scaling ensures that numerical values are standardized so the model can learn efficiently.

7. Data Splitting

The dataset is divided into training and testing datasets.

Data Split

Dataset	Percentage	Purpose
Training Data	80%	Used to train the model
Testing Data	20%	Used to evaluate model performance

Testing Dataset Size

Parameter	Value
Test Transactions	20,000
Normal Transactions	19,800
Fraud Transactions	200

8. Machine Learning Model

The project uses the **Random Forest Classifier**.

Random Forest is an ensemble machine learning algorithm that builds multiple decision trees and combines their predictions to improve accuracy.

Random Forest Advantages

Advantage	Description
High Accuracy	Produces reliable predictions
Handles Large Data	Works well with large datasets
Reduces Overfitting	Uses multiple decision trees
Robust Model	Suitable for classification problems

The model is trained using **300 decision trees**.

9. Model Predictions

After training, the model predicts whether each transaction in the test dataset is **normal or fraudulent**.

Example Predictions

Transaction	Prediction
Transaction 1	Normal
Transaction 2	Normal
Transaction 3	Normal
Transaction 4	Normal
Transaction 5	Normal

10. Prediction Summary

From the testing dataset of **20,000 transactions**, the model predicted:

Prediction Type	Count
Fraud Predictions	1,663
Normal Predictions	18,337

11. Confusion Matrix

The confusion matrix shows the detailed prediction performance of the model.

Confusion Matrix Result

	Predicted Normal	Predicted Fraud
Actual Normal	18,151	1,649
Actual Fraud	186	14

Explanation

Term	Meaning
True Negative (18151)	Correctly predicted normal transactions
False Positive (1649)	Normal transactions predicted as fraud
False Negative (186)	Fraud transactions predicted as normal
True Positive (14)	Correctly detected fraud transactions

12. Model Accuracy

The overall accuracy of the model is:

Accuracy

Metric	Value
Model Accuracy	**90.82%**

This means the model correctly predicted about **90% of transactions**.

13. Classification Report

The classification report provides more detailed performance metrics.

Model Performance Metrics

Class	Precision	Recall	F1 Score	Support
Normal (0)	0.99	0.92	0.95	19,800
Fraud (1)	0.01	0.07	0.02	200

Metric Explanation

Metric	Meaning
Precision	How many predicted fraud transactions were correct
Recall	Ability of the model to detect fraud transactions
F1 Score	Balance between precision and recall
Support	Number of samples in the dataset

The model performs very well for normal transactions but still struggles to detect fraud accurately.

14. Conclusion

This project successfully demonstrates the implementation of a **Credit Card Fraud Detection System using Machine Learning**.

The system analyses transaction data, balances the dataset using SMOTE, and uses a Random Forest classifier to detect fraudulent transactions.

The model achieved an overall accuracy of **90.82%**, showing good performance in identifying normal transactions. However, fraud detection can still be improved by using more advanced models or additional features.

With further improvements and real-time implementation, this system can help financial institutions detect fraudulent transactions and reduce financial losses